REAR PRE-TENSIONER SEAT BELT [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

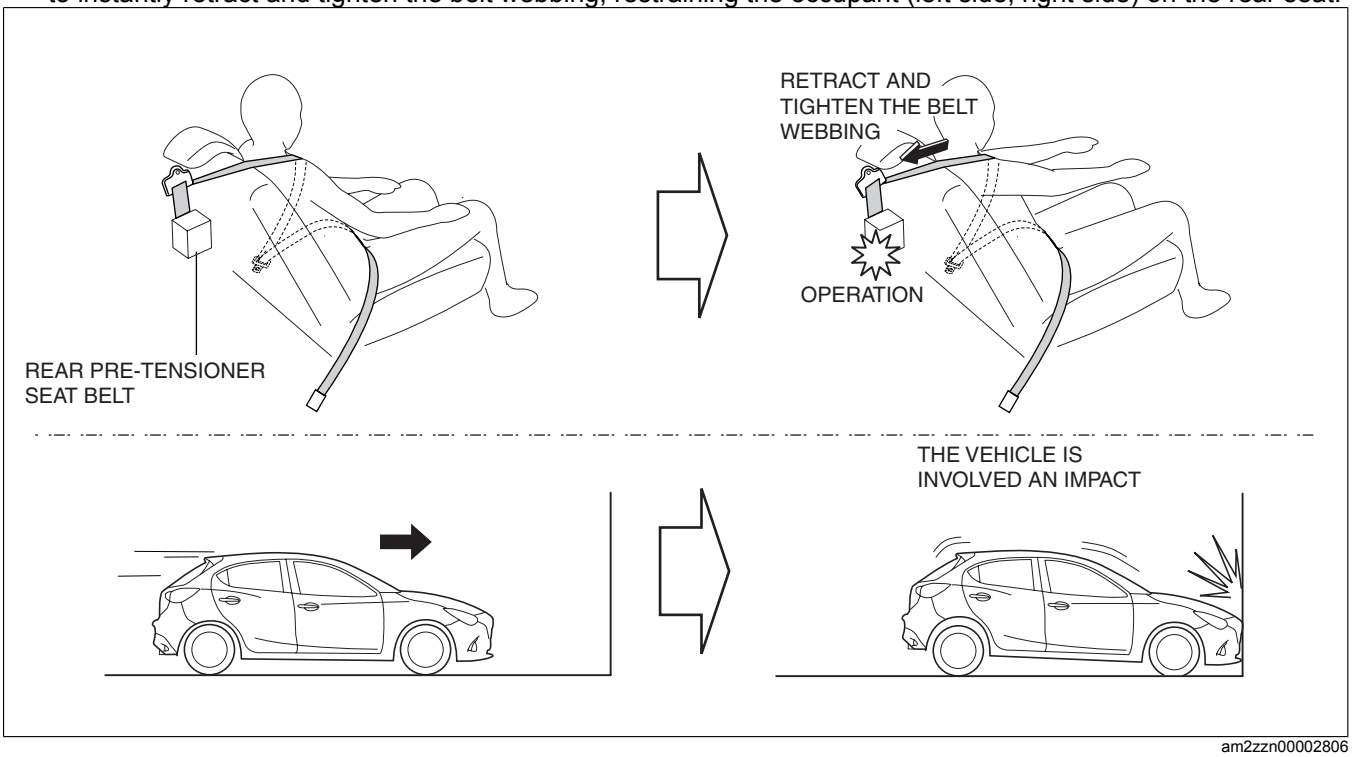
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Purpose

- The rear pre-tensioner seat belt retracts and tightens the belt webbing (between rear buckle and rear pre-tensioner seat belt) to protect the occupant during a frontal or front-sideward collision.

Function

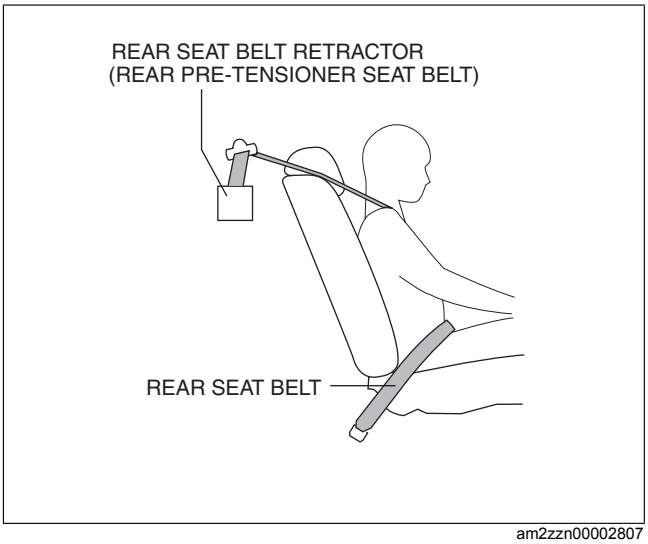
- The rear pre-tensioner seat belt operates (deploys) based on the operation signal from the SAS control module to instantly retract and tighten the belt webbing, restraining the occupant (left side, right side) on the rear seat.



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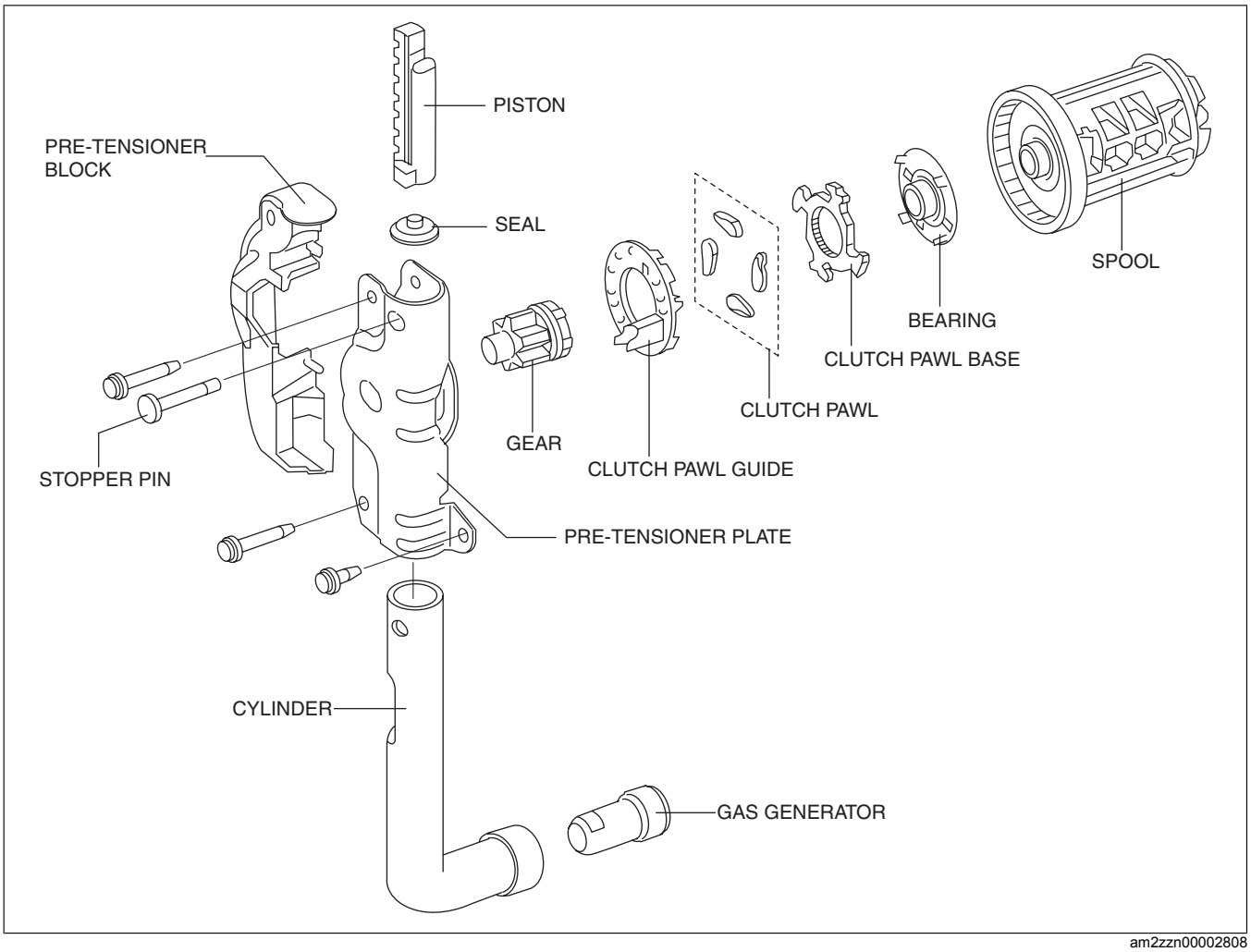
Construction

- The rear pre-tensioner seat belt is installed to the retractor of the rear seat belt (left side, right side).



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- The rear pre-tensioner seat belt consists of the following parts shown in the figure:

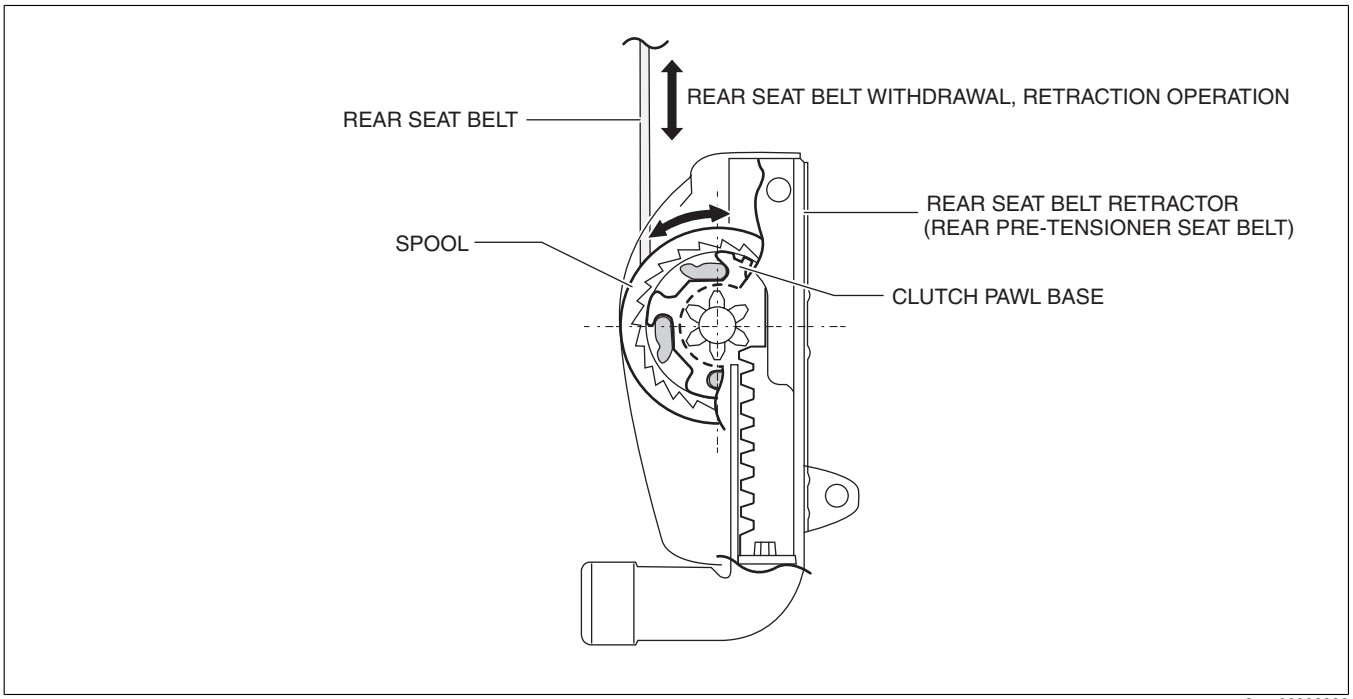


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Operation

Normal operation (pre-tensioner non-operation)

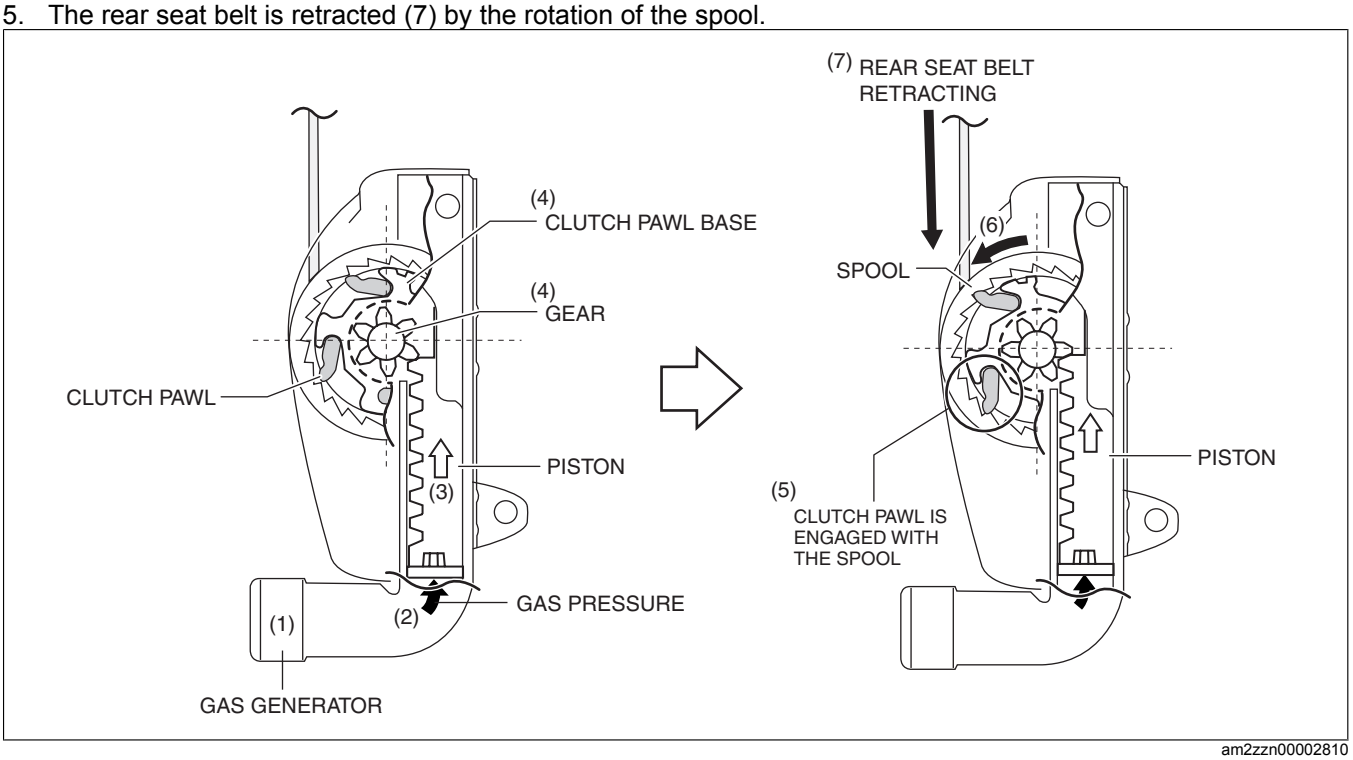
- The spool (retraction shaft) of the rear pre-tensioner seat belt is separated from the clutch pawl base.
- The spool of the rear pre-tensioner seat belt rotates in conjunction with the with drawl/retraction of the rear seat belt.



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Pre-tensioner activation

1. The gas generator produces (1) gas when it receives an operation signal from the SAS control module.
2. The piston in the cylinder is pressed (3) by the generated gas pressure (2) and the gear and the clutch pawl base rotate (4).
3. The clutch pawl is pulled out and engaged (5) with the spool by the rotation of the gear and the clutch pawl base. The ring gear is engaged with the locking base and the force of the ball being pressed out is transmitted to the retraction shaft.
4. When the clutch pawl is engaged, the force of the pressed piston is transmitted to the spool and the spool rotates (6).
5. The rear seat belt is retracted (7) by the rotation of the spool.



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Fail-safe

- Not applicable